

Performance Analysis of Multi-User NOMA with Adaptive Modulation over Multi-Fading Channels

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Abstract

Non-Orthogonal Multiple Access (NOMA) is an important multiple access technique used in modern wireless communication systems to improve spectral efficiency and support multiple users simultaneously. In this work, the performance of a NOMA system with adaptive modulation is analyzed under different wireless channel conditions. Unlike conventional approaches that consider only a single fading environment, this work evaluates the system performance over three different fading channels: Rayleigh, Rician, and Nakagami, which represent various real-world communication scenarios. The system is further extended to support multiple users using superposition coding at the transmitter and successive interference cancellation (SIC) at the receiver for signal detection. In addition, a dynamic power allocation strategy is implemented where transmission power is allocated to users based on their channel conditions, providing higher power to users with weaker channel gains to improve fairness and reliability. Adaptive modulation techniques are also incorporated to adjust the modulation scheme according to the signal-to-noise ratio (SNR), thereby enhancing overall system performance. The proposed system is implemented and simulated in MATLAB, and its performance is evaluated in terms of Bit Error Rate (BER) with respect to SNR. The simulation results help analyze the impact of different fading channels, adaptive modulation, and dynamic power allocation on the efficiency and reliability of the NOMA communication system.

Keywords: Non-Orthogonal Multiple Access (NOMA); Adaptive Modulation; Fading Channels ; Bit Error Rate (BER); Successive Interference Cancellation (SIC).

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Abbreviations

Abbreviation	Description
NOMA	Non-Orthogonal Multiple Access
OFDM	Orthogonal Frequency Division Multiplexing
SIC	Successive Interference Cancellation
BER	Bit Error rate
SNR	Signal-to-Noise ratio
BPSK	Binary phase shift keying
QPSK	Quadrature phase shift keying
QAM	Quadrature Amplitude Modulation

CHAPTER 1

Introduction

1.1 Introduction

Modern wireless communication systems demand high data rates, improved spectral efficiency, and the ability to support multiple users within limited bandwidth resources. Traditional multiple access techniques such as Time Division Multiple Access (TDMA) and Frequency Division Multiple Access (FDMA) allocate separate resources to each user, which limits the overall system efficiency and user capacity. To overcome these limitations, Non-Orthogonal Multiple Access (NOMA) has emerged as an effective multiple access technique for next-generation wireless communication systems. In NOMA, multiple users share the same time and frequency resources by assigning different power levels, and the transmitted signals are combined using superposition coding. At the receiver side, Successive Interference Cancellation (SIC) is used to separate and decode the signals of different users.

However, the performance of NOMA systems is highly influenced by channel conditions such as noise and fading. In practical wireless environments, signals experience various types of fading due to multipath propagation and obstacles. Therefore, it is important to analyze the system under different fading channel models such as Rayleigh, Rician, and Nakagami, which represent different real-world propagation scenarios. In this project, a multi-user NOMA system is analyzed by incorporating adaptive modulation and dynamic power allocation techniques. Adaptive modulation allows the system to select suitable modulation schemes such as BPSK and QPSK based on the Signal-to-Noise Ratio (SNR), while dynamic power allocation assigns transmission power based on user channel conditions to improve fairness and reliability. The system is implemented and simulated using MATLAB, and its performance is evaluated in terms of Bit Error Rate (BER) versus SNR. The results provide insights into

improving the efficiency and reliability of multi-user wireless communication systems under different channel environments.

1.2 Background and Motivation

Wireless communication systems have grown rapidly in recent years due to the increasing demand for high data rates, reliable connectivity, and efficient use of limited spectrum resources. Traditional multiple access techniques, such as Time Division Multiple Access (TDMA) and Frequency Division Multiple Access (FDMA), allocate separate resources like time slots or frequency bands, to different users. Although these methods reduce interference among users, they limit the number of users that can be supported simultaneously and reduce spectral efficiency. To overcome these limitations, Non-Orthogonal Multiple Access (NOMA) has been introduced as an efficient multiple access technique for modern wireless communication systems. In NOMA, multiple users share the same time and frequency resources by allocating different power levels based on their channel conditions. The signals of different users are combined using superposition coding at the transmitter, and at the receiver side Successive Interference Cancellation (SIC) is used to separate and detect the signals.

The performance of wireless communication systems is significantly affected by channel fading and noise. In real-world environments, transmitted signals experience different types of fading due to multipath propagation and obstacles in the communication path. Common fading models used in wireless communication include Rayleigh, Rician, and Nakagami fading channels, each representing different propagation conditions such as non-line-of-sight and line-of-sight environments. Therefore, analyzing the performance of NOMA systems under these different fading conditions is important for understanding system reliability and efficiency. In addition, modern communication systems require adaptive techniques that can adjust transmission parameters according to changing channel conditions.

The motivation for this project is to improve the performance and reliability of multi-user wireless communication systems by incorporating adaptive

techniques within a NOMA framework. In this work, the NOMA system is extended to support multiple users, and adaptive modulation schemes such as BPSK and QPSK are used to improve communication reliability under varying signal-to-noise ratio (SNR) conditions. Furthermore, a dynamic power allocation strategy is implemented in which transmission power is assigned to users based on their channel conditions, allowing users with weaker channels to receive higher power. By analyzing the system under Rayleigh, Rician, and Nakagami fading channels using MATLAB simulations, the project aims to evaluate the impact of different channel environments, adaptive modulation, and dynamic power allocation on the Bit Error Rate (BER) performance of the NOMA system. This study helps provide insights into improving spectral efficiency, fairness among users, and overall system performance in modern wireless communication networks.

1.3 Objectives of the Project Work

- To develop and simulate a NOMA-based wireless system using MATLAB for efficient multi-user transmission
- To analyze system performance under different fading channel models such as Rayleigh, Rician, and Nakagami.
- To implement adaptive modulation techniques to enhance performance under varying SNR conditions.
- To implement dynamic power allocation strategies among multiple users based on channel conditions to improve fairness and system throughput.
- To evaluate and compare the Bit Error Rate (BER) and spectral efficiency of the system for different user configurations and channel scenarios.

CHAPTER 2

Literature Survey

2.1 Introduction

In the context of this work, the literature survey focuses on research related to Non-Orthogonal Multiple Access (NOMA), adaptive modulation techniques, multi-user communication systems, and the impact of different wireless fading channels such as Rayleigh, Rician, and Nakagami. Many researchers have studied the performance of NOMA systems and proposed various techniques to improve spectral efficiency, reliability, and user fairness in modern wireless communication networks. These studies provide valuable insights into power allocation strategies, interference management using Successive Interference Cancellation (SIC), and the role of adaptive techniques in improving system performance.

The literature reviewed in this chapter is collected from various research papers, journals, and conference publications related to wireless communication and multiple access technologies. The objective of reviewing these works is to understand the existing methods, compare different approaches, and identify the research gap that motivates the proposed project. This chapter presents a summary of selected research papers that discuss NOMA systems, adaptive modulation methods, and performance analysis under different fading channel conditions. The key contributions and findings of these studies are discussed to provide a clear background for the proposed work.

2.2 Review of Prior Research

Previous studies in the field of Non-Orthogonal Multiple Access (NOMA) have primarily focused on improving spectral efficiency and enabling multiple users to share the same communication resources effectively. A significant amount of research has been dedicated to enhancing signal detection techniques,

especially through the use of Successive Interference Cancellation (SIC). These works demonstrate that efficient interference management is essential for improving system reliability and reducing error rates in multi-user communication environments. In a typical SIC process, the signal with the highest power is detected first, and its contribution is removed from the received signal before detecting the remaining users. However, this approach is highly sensitive to detection errors. If the stronger signal is incorrectly detected, the error propagates to subsequent stages, significantly degrading the performance of the overall system. Additionally, reliable operation of conventional SIC requires a large difference in power levels between users, which limits system flexibility.

One of the literature reviews that we considered is **Improved Detection in Successive Interference Cancellation NOMA OFDM Receiver** by Krzysztof W. Wesolowski[1] had worked on Non-Orthogonal Multiple Access (NOMA), which has gained significant attention in modern wireless communication systems due to its ability to improve spectral efficiency and support multiple users within limited radio resources. Unlike conventional orthogonal multiple access techniques, NOMA allows multiple users to share the same time and frequency resources by superimposing their signals with different power levels. At the receiver, Successive Interference Cancellation (SIC) is employed to separate and decode these signals. The performance of NOMA systems largely depends on the effectiveness of the SIC receiver, making it a critical component in such communication systems. Another literature review that I referred to is **Enhanced BER-Analysis-Based Adaptive Resource Allocation in SDN-Controlled Multi-User NOMA-OFDM VLC Systems** by Chengju Hu, Jianhang Li, Yinan Niu, and Jian Zhao [2] proposed a novel approach to improve the performance and fairness of visible light communication (VLC) systems serving multiple users. By integrating non-orthogonal multiple access (NOMA) with OFDM and leveraging a software-defined network (SDN) controller, the study introduces an adaptive resource allocation algorithm based on bit-error-rate (BER) analysis. The algorithm dynamically optimizes power and subcarrier allocation according to channel conditions, addressing the limitations of conventional static allocation methods. Through

successive convex approximation and block coordinate descent, the method efficiently solves the underlying non-convex optimization problem. Simulation and experimental results demonstrate significant improvements in system throughput, BER performance, and user fairness, making it a promising solution for next-generation high-speed VLC networks.

2.3 Summary

The reviewed works collectively focus on improving the performance and efficiency of multi-user Non-Orthogonal Multiple Access (NOMA) systems in modern communication networks. These approaches highlight the significance of advanced signal processing and adaptive optimization techniques to overcome the limitations of traditional orthogonal multiple access methods, especially in scenarios with limited spectral resources and increasing user demand.

A key aspect emphasized is the role of Successive Interference Cancellation (SIC) at the receiver. Since NOMA relies on power-domain multiplexing where multiple user signals are superimposed, accurate signal separation becomes critical. Enhancements in detection techniques within the SIC framework help reduce error propagation, improve decoding accuracy, and ensure reliable communication. This ultimately contributes to better spectral efficiency and system performance.

Another important focus is on adaptive resource allocation mechanisms, where system parameters such as power distribution and subcarrier assignment are dynamically adjusted based on channel conditions and performance metrics like Bit Error Rate (BER). By replacing static allocation strategies with intelligent and flexible methods, the system can achieve improved throughput, reduced error rates, and better fairness among users.

Additionally, the integration of advanced optimization techniques enables efficient handling of complex system models, allowing for real-time adaptation in dynamic environments. These approaches are particularly beneficial in emerging communication scenarios, including optical and wireless hybrid systems.

Overall, the combined insights demonstrate that efficient interference management at the receiver and adaptive resource optimization at the system level

are both essential for enhancing the performance, reliability, and scalability of multi-user NOMA systems, making them highly suitable for next-generation communication technologies.

CHAPTER 3

NOMA-OFDM System Analysis for Rayleigh Fading channel

3.1 Introduction

With the rapid growth of wireless communication and the increasing demand for high data rates, better spectral efficiency, and massive connectivity, advanced multiple access techniques have become essential. One such promising technique is Non-Orthogonal Multiple Access (NOMA), which allows multiple users to share the same time and frequency resources by allocating different power levels. This improves spectrum utilization compared to traditional orthogonal schemes.

On the other hand, Orthogonal Frequency Division Multiplexing (OFDM) is a widely used modulation technique in modern wireless systems due to its robustness against frequency-selective fading and its ability to efficiently handle multipath propagation. By combining NOMA with OFDM, the resulting NOMA-OFDM system leverages the advantages of both technologies—enhanced spectral efficiency from NOMA and resilience to channel impairments from OFDM.

In practical wireless environments, signals are often affected by multipath fading, where the transmitted signal experiences random amplitude variations. A commonly used model to represent such conditions is the Rayleigh fading channel, which assumes no direct line-of-sight path between transmitter and receiver. Analyzing NOMA-OFDM systems under Rayleigh fading conditions is crucial for understanding real-world performance, including metrics such as bit error rate (BER), outage probability, and system capacity.

This study focuses on evaluating the performance of NOMA-OFDM systems in Rayleigh fading environments, highlighting the impact of channel conditions, power allocation strategies, and user pairing on overall system efficiency and

reliability.

3.2 Problem addressed

The key issue identified in the [1] project is that the performance of conventional Successive Interference Cancellation (SIC) receivers in Non-Orthogonal Multiple Access (NOMA) systems is significantly affected by inaccurate signal detection, which leads to error propagation. In a typical SIC process, the receiver detects the strongest user signal first and then subtracts it from the received signal to detect the remaining users. However, if the detection of the stronger signal is incorrect, the error propagates to subsequent stages, resulting in incorrect detection of weaker user signals and overall degradation of system performance.

Additionally, the conventional SIC approach assumes that weaker user signals can be treated as noise during the detection of stronger signals. In reality, these signals are structured and interfere with accurate detection, especially when the power difference between users is small. This makes the system less reliable and limits its practical applicability. Therefore, the paper aims to address this issue by improving the detection process in SIC receivers, reducing error propagation, and enhancing detection accuracy even under conditions where users have similar power levels.

The key issue identified in another work [2] is the inefficient and unfair resource allocation in multi-user VLC systems using NOMA-OFDM under dynamic channel conditions. In conventional schemes, power and subcarrier allocation are often fixed or based on simple rules, which cannot adapt to changing channel conditions in visible light communication. This leads to higher bit error rates (BER) for some users, especially those with weaker channels (far users), reducing overall system performance and fairness. Additionally, static allocation fails to exploit the full potential of multi-user NOMA in OFDM-based VLC systems, limiting throughput.

3.3 Existing Method

In [1], they proposed an improved detection method for the Successive Interference Cancellation (SIC) receiver to overcome the problem of error propagation in NOMA systems. Instead of treating weaker user signals as noise, the method makes tentative decisions about these signals and uses them to assist in the detection of stronger user signals. This approach improves the accuracy of detecting the strongest signal in the initial stage. Once the stronger signal is correctly detected, it is subtracted, leading to more reliable detection of the remaining weaker signals.

In [2], the proposed method introduces an adaptive resource allocation strategy for SDN-controlled multi-user NOMA-OFDM VLC systems, where power and OFDM subcarriers are dynamically assigned to users based on their channel conditions. Users with weaker channels are allocated higher power or better subcarriers, while users with stronger channels receive comparatively less, thereby optimizing overall system performance in terms of BER and throughput. The approach focuses on minimizing the Bit Error Rate (BER) while ensuring fairness among all users by incorporating suitable constraints so that reliable communication is maintained for each user. To efficiently solve the resulting non-convex optimization problem, iterative techniques such as Successive Convex Approximation (SCA) and Block Coordinate Descent (BCD) are employed, enabling near-optimal resource allocation. Furthermore, the integration of a Software-Defined Network (SDN) controller allows continuous monitoring of channel conditions and real-time updating of power and subcarrier assignments, leading to improved adaptability, enhanced performance, and better fairness in multi-user VLC systems.

3.4 System Model

The system considers an **uplink NOMA-OFDM communication system** in which multiple users transmit their signals simultaneously over the same frequency band, and the signals are **superimposed at the receiver** with

different power levels. The receiver employs Successive Interference Cancellation (SIC) to detect and separate the user signals step by step, where the strongest signal is detected first and then subtracted to decode the weaker signals. The system assumes a limited number of users sharing common radio resources and also accounts for the presence of noise and interference affecting signal detection. The proposed improved detection method is incorporated within this model to enhance detection accuracy and overall system performance.

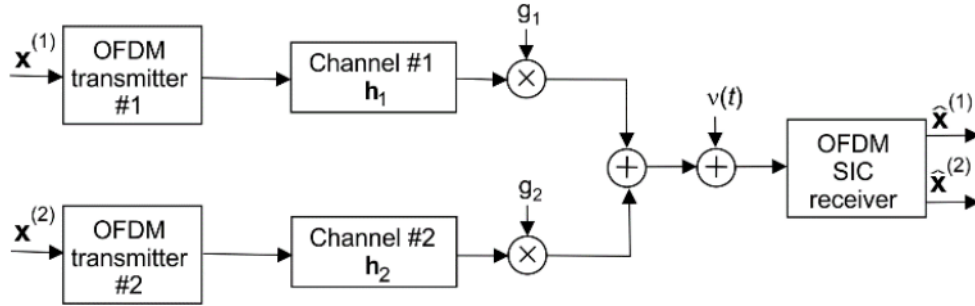


Figure 3.1: Uplink NOMA-OFDM system

The signal on each OFDM subcarrier output can be expressed by a simple equation:

$$Y_k = H_k^{(1)} X_k^{(1)} + H_k^{(2)} X_k^{(2)} + N_k$$

The model that is generated in [2] considers a downlink VLC network where a single LED transmitter serves multiple users at different distances, resulting in varying channel gains. The system employs **multi-user NOMA combined with OFDM**, allowing multiple users to share the same subcarrier with different power levels—far users receive higher power while near users receive lower power to ensure fairness. The VLC channel is modeled as line-of-sight (LOS) dominant with Lambertian radiation patterns, and noise is considered as additive white Gaussian noise (AWGN). A **software-defined network (SDN)** controller monitors user channel conditions and dynamically allocates power and subcarriers based on predicted bit-error-rate (BER), enabling real-time adaptive resource allocation. The main objective of the system is to minimize total BER across all users while satisfying transmitter power constraints, with the non-convex optimization problem solved iteratively using successive convex

approximation (SCA) and block coordinate descent (BCD) techniques.

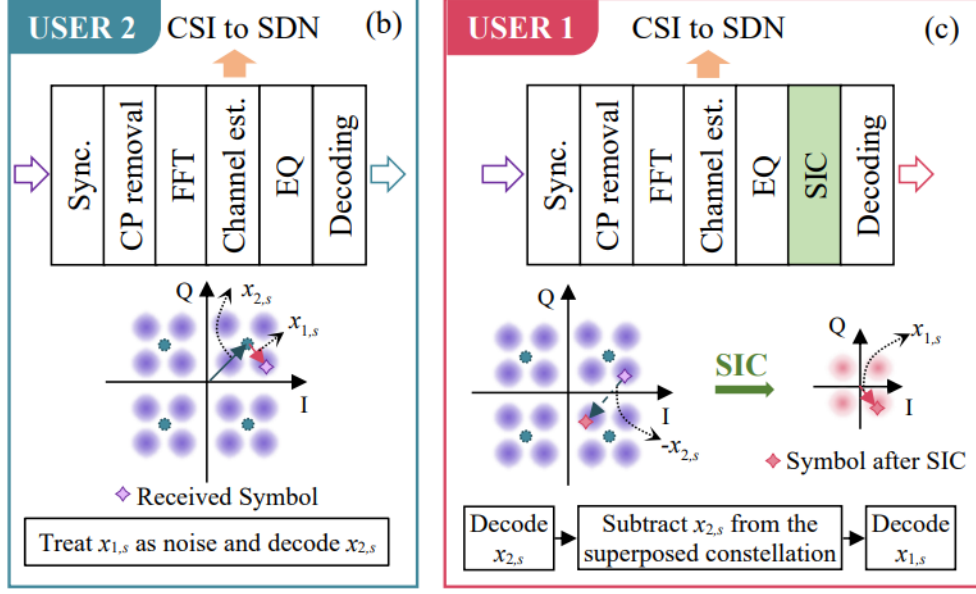


Figure 3.2: Two-user NOMA-OFDM system

3.5 Limitations

- Lack of analysis under multiple fading channels like Rayleigh, Rician, and Nakagami:** The detailed evaluation of system performance under different wireless fading environments is not provided. In practical communication systems, signals experience various types of fading depending on the environment. Without analyzing multiple channel models, it is difficult to understand how the system performs in real-world scenarios. Hence, studying different fading channels is necessary to evaluate system reliability and robustness.
- Limited to a small number of users, no multi-user scalability analysis:** Considers a limited number of users in the NOMA system. However, modern wireless communication systems require support for multiple users simultaneously. The impact of increasing the number of users on system performance, interference, and reliability is not thoroughly analyzed. Studying multi-user scenarios is essential for practical implementation in real-world networks.

- **Limited to only one modulation technique for varying channel conditions:** The study does not incorporate adaptive modulation schemes that adjust according to the signal-to-noise ratio (SNR). Using a fixed modulation scheme may not provide optimal performance under all channel conditions. Adaptive modulation helps improve system reliability at low SNR and increases data rate at high SNR. Hence, integrating adaptive modulation is important for enhancing system performance.
- **Limited integration of channel, modulation, and power optimization together:** The work does not combine multiple optimization techniques, such as channel modeling, adaptive modulation, and dynamic power allocation, into a single framework. Each of these factors plays a crucial role in communication system performance. A combined approach can significantly improve efficiency, reliability, and spectral utilization. Therefore, there is a need for a more integrated system design, which is addressed in the proposed project.

Both the approaches significantly enhance the performance of NOMA-based communication systems by addressing key limitations of conventional techniques. The improved detection method for the Successive Interference Cancellation (SIC) receiver effectively mitigates error propagation by utilizing tentative decisions, resulting in more accurate signal detection and reliable decoding of both strong and weak user signals. Meanwhile, the adaptive resource allocation strategy in SDN-controlled multi-user NOMA-OFDM VLC systems optimizes power and subcarrier distribution based on channel conditions, thereby reducing BER and improving throughput while maintaining fairness among users. The use of advanced optimization techniques such as SCA and BCD, along with real-time control through SDN, further strengthens system adaptability and efficiency. Overall, these methods contribute to more robust, efficient, and intelligent multi-user communication systems, making them well-suited for next-generation wireless and optical networks.

3.6 Summary

Both papers focus on improving multi-user NOMA-OFDM systems, but from complementary perspectives address resource allocation in VLC networks, proposing a BER-analysis-based adaptive algorithm integrated with an SDN controller to dynamically assign power and subcarriers. This approach minimizes BER, ensures fairness, and increases throughput, particularly benefiting far users, with simulations and experiments confirming superior performance compared to static allocation methods.

The [1] focuses on the receiver side of RF NOMA-OFDM systems, enhancing the successive interference cancellation (SIC) process. It introduces advanced detection techniques to reduce error propagation and improve decoding accuracy for multi-user signals, resulting in lower BER and better system reliability even under challenging interference conditions.

The [2] uses adaptive resource allocation in a multi-user NOMA-OFDM VLC system, where power and subcarriers are assigned based on channel conditions. It minimizes BER, ensures fairness among users, and uses optimization techniques with SDN control for efficient and dynamic system performance. Together, these studies provide a holistic view of NOMA-OFDM optimization, with the first improving transmission and resource allocation and the second enhancing receiver detection and interference management, offering strategies to maximize both performances and fairness in multi-user communication systems.

CHAPTER 4

Multi-User NOMA Performance over Fading Channels

4.1 Introduction

The project investigates the performance of multi-user NOMA-OFDM systems under realistic wireless environments by considering multiple fading channels, including Rayleigh, Rician, and Nakagami. It extends prior work on successive interference cancellation (SIC) receivers by analyzing system-level performance with adaptive modulation techniques such as BPSK and QPSK. By dynamically adjusting modulation and power allocation based on channel conditions, the project aims to minimize bit error rate (BER), enhance throughput, and ensure fairness among users. The study provides insights into the robustness and efficiency of multi-user NOMA systems in varying channel conditions, making it relevant for next-generation wireless communication networks.

4.2 System Model

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This system begins with user data streams being processed through adaptive modulation, which dynamically selects the optimal modulation scheme based on channel conditions to maximize spectral efficiency. The NOMA encoder then applies superposition coding, allocating different power levels to multiple users—higher power to users with poor channel conditions and lower power to those with better channels—enabling simultaneous transmission on the same time-frequency resources. This composite signal undergoes OFDM modulation, converting it into parallel subcarriers to combat multi-path fading before transmission over multi-fading channels (h , h , etc.), where signals expe-

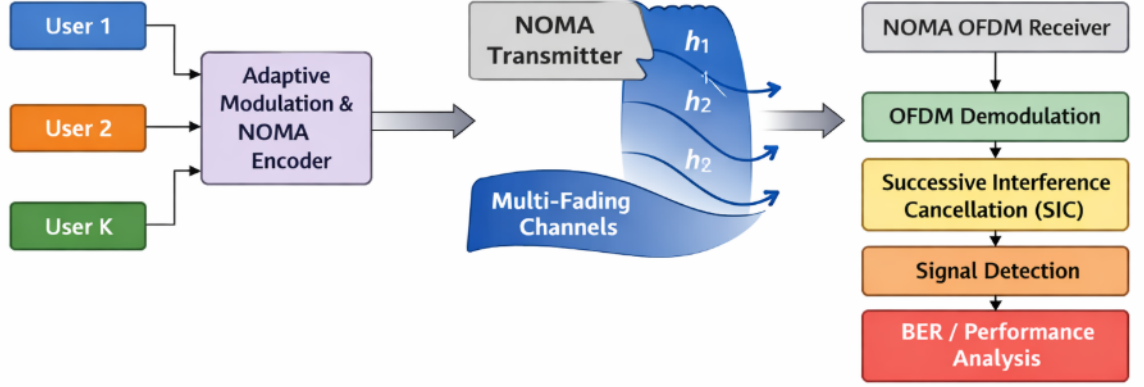


Figure 4.1: Multi-user NOMA system

rience independent fading effects. At the receiver side, OFDM demodulation reconstructs the subcarriers, followed by Successive Interference Cancellation (SIC), which decodes users sequentially by first demodulating and subtracting the high-power signal (the weak user) before extracting the low-power signal (the strong user). Finally, signal detection is performed to recover the original user data, and the system evaluates performance through Bit Error Rate (BER) analysis, providing critical insights into the trade-offs between power allocation, user fairness, and overall system reliability under realistic fading conditions.

4.3 Methodology

The proposed system implements a **multi-user NOMA (Non-Orthogonal Multiple Access)** wireless communication framework designed to support three users simultaneously with differing channel conditions and modulation schemes. The methodology consists of the following key components:

1. **User Configuration and Power Allocation:** The system considers three users positioned at different distances from the transmitter. Power allocation is performed according to user location, where the far user receives the highest power, the mid-range user receives moderate power, and the near user receives the lowest power. This ensures fairness and reliable communication for all users.
2. **Adaptive Modulation:** Each user employs a modulation scheme suited

to their channel and power allocation. The far user uses **BPSK**, the mid-range user uses **QPSK**, and the near user uses **16-QAM**. Adaptive modulation balances spectral efficiency and error performance.

3. **Channel Modeling:** The system is simulated over three fading channels to represent realistic wireless environments:

- **Rayleigh fading:** Non-line-of-sight channel conditions.
- **Rician fading:** Dominant line-of-sight component.
- **Nakagami-m fading:** Flexible fading with varying severity.

4. **Successive Interference Cancellation (SIC):** SIC is applied at the receiver to sequentially decode user signals. The far user is decoded first due to higher power allocation, followed by the mid-range and near users. SIC mitigates inter-user interference, enabling simultaneous transmission.

5. **Performance Evaluation:** The system performance is evaluated using **Bit Error Rate (BER)** as a function of **Signal-to-Noise Ratio (SNR)**. MATLAB simulations analyze the effect of fading channels, adaptive modulation, and power allocation on system reliability and efficiency.

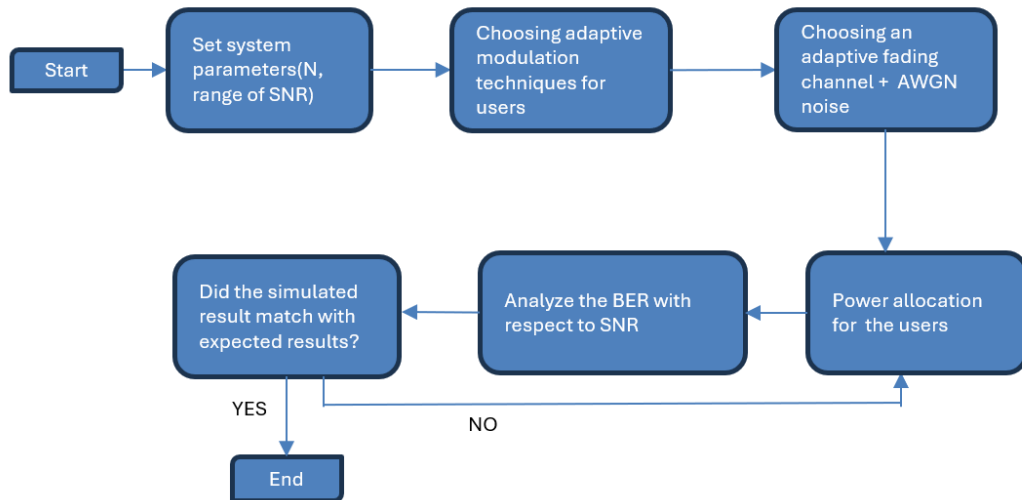


Figure 4.2: Methodology

This methodology represents a typical simulation workflow for evaluating a communication system under fading channel conditions with adaptive techniques

in MATLAB. The process begins with setting the system parameters such as the number of users (N) and the range of Signal-to-Noise Ratio (SNR) values. These parameters define the operating environment for the simulation and help in analyzing system performance across different channel conditions.

Next, adaptive modulation techniques are selected for the users based on the channel conditions. In parallel, an adaptive fading channel model is chosen along with the addition of AWGN (Additive White Gaussian Noise) to realistically simulate wireless communication impairments. Based on the channel quality and user requirements, power allocation is performed among the users to optimize overall system performance, especially in multi-user scenarios like NOMA.

After configuring the system and channel parameters, the performance is evaluated by analyzing the Bit Error Rate (BER) with respect to SNR. This step involves running simulations and observing how BER varies as SNR changes, which helps in understanding the reliability of the communication system under different conditions. The adaptive nature of modulation and power allocation plays a key role in improving performance.

Finally, the simulated results are compared with the expected theoretical or benchmark results. If the results match within acceptable limits, the process is concluded; otherwise, adjustments are made to system parameters, modulation schemes, or power allocation strategies, and the simulation is repeated. This iterative validation ensures the accuracy and effectiveness of the proposed system design.

4.4 Summary

This project implements a multi-user NOMA (Non-Orthogonal Multiple Access) communication system with adaptive modulation over Rayleigh, Rician, and Nakagami fading channels using MATLAB. Three users are considered: a far user with BPSK, a mid-range user with QPSK, and a near user with 16-QAM, with power allocated according to their channel conditions. The system performs superposition coding at the transmitter and successive interference cancellation (SIC) at the receivers to detect signals in a hierarchical

manner. The performance of each user is evaluated in terms of bit error rate (BER) across varying SNR levels for different channel types. The simulation demonstrates the effectiveness of NOMA with SIC in multi-user scenarios, highlighting how power allocation and channel conditions influence BER, and allows comparison of fading channels for system reliability. Graphical results illustrate BER trends for individual users and across all channels, providing insights into practical NOMA system design.

CHAPTER 5

Results and Discussion

5.1 Introduction

In this section, the performance of the proposed multi-user NOMA system is analyzed in terms of bit error rate (BER) under different fading channels and modulation schemes. The system considers three users with BPSK, QPSK, and 16-QAM modulation and employs successive interference cancellation (SIC) at the receivers to detect signals efficiently. Simulations are carried out over Rayleigh, Rician, and Nakagami fading channels, with varying signal-to-noise ratios (SNRs) from 0 dB to 20 dB. The discussion focuses on the impact of channel conditions, power allocation, and modulation order on system performance. BER trends are presented graphically for individual users and compared across channel types, highlighting how NOMA with SIC enables simultaneous transmission for multiple users while maintaining acceptable error rates. This analysis provides insights into practical system design and identifies the scenarios in which each user and channel type performs optimally.

5.2 Overview of Results

The proposed multi-user NOMA system employs adaptive modulation and successive interference cancellation (SIC) to allow multiple users to share the same frequency band simultaneously. Each user is assigned a different power level based on their distance from the transmitter, with far users receiving higher power and near users lower power. The system performance depends on factors such as channel conditions, modulation order, and power allocation. BER is used as the primary metric to evaluate reliability, and different fading channels such as Rayleigh, Rician, and Nakagami are considered to model realistic wireless environments.

5.3 Analysis and Interpretation

5.3.1 User-wise BER Performance

- **User 1 (BPSK, far user, highest power allocation 0.7)** consistently achieves the **lowest BER** across all channels due to higher transmission power and robust modulation.
- **User 2 (QPSK, mid-range power 0.2)** exhibits moderate BER, benefiting from successive interference cancellation (SIC) that cancels interference from User 1.
- **User 3 (16-QAM, near user, lowest power 0.1)** experiences the **highest BER** because of lower power allocation and higher-order modulation, making it more sensitive to noise and residual interference.

5.3.2 Channel Comparison

- **Rician channel ($K=5$)** shows slightly better BER than **Rayleigh**, particularly for the far user, due to the presence of a strong line-of-sight component.
- **Nakagami-m fading** with $m > 1$ improves BER for closer users compared to Rayleigh, highlighting how the fading parameter influences performance.
- Overall, **User 1 performs reliably across all channels**, whereas User 3's performance is most affected by channel variations.

5.3.3 Effect of SNR

- BER decreases with increasing SNR for all users and channels, confirming the effectiveness of NOMA with SIC in mitigating interference.
- High SNR values lead to **near-error-free performance for User 1**, while User 3 still shows some errors at moderate SNRs due to higher-order modulation and lower power.

5.3.4 SIC Efficiency

- Successive interference cancellation successfully isolates user signals, enabling simultaneous transmission.

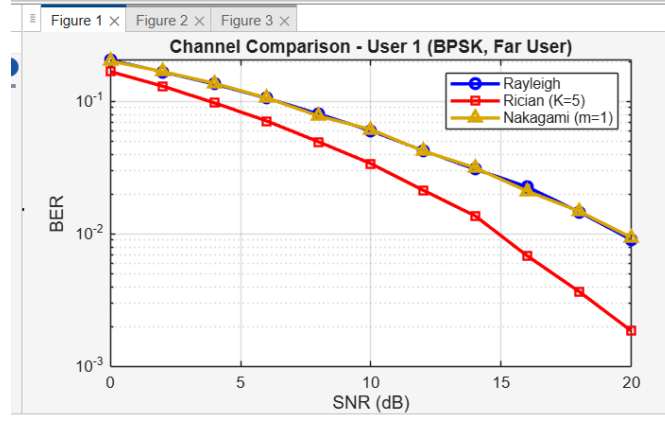


Figure 5.1: BER performance for User 1 across different channels

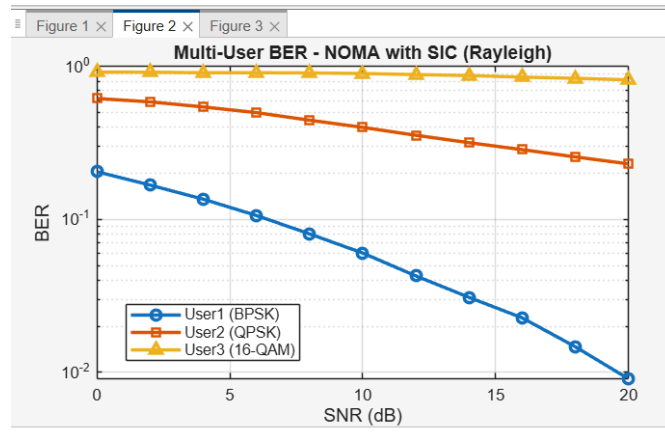


Figure 5.2: Multi-user BER performance under Rayleigh fading

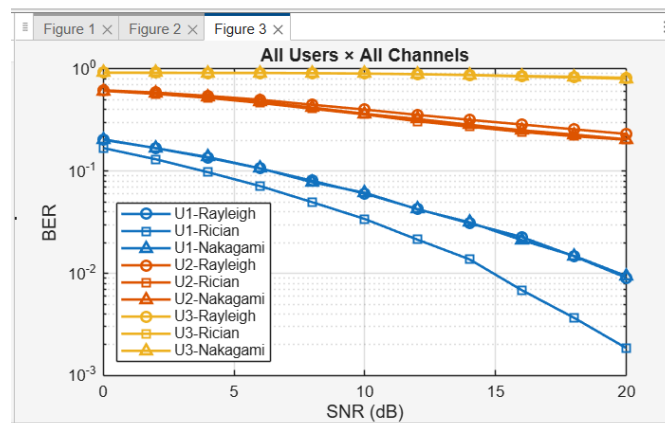


Figure 5.3: BER comparison of all users across all channels

Table 5.1: Comparison Between Existing and Proposed Work

Existing work	Proposed
SIC Detection Improvement	Performance analysis over multi-fading channels
NOMA-OFDM wireless system	NOMA-OFDM wireless system
Improved SIC with tentative decisions	Power and subcarrier allocation
Reduced error propagation	Minimized BER and improved fairness
Detection accuracy	BER vs SNR analysis
BER ($\geq 10^{-1}$)	Enhanced BER rate performance ($< 10^{-1}$)

The simulation results illustrate the performance of a multi-user NOMA system with Successive Interference Cancellation (SIC) under different modulation schemes and fading channel conditions.

Figure 5.1 (Multi-user BER with Rayleigh channel), it is observed that User 1 (BPSK) achieves the lowest Bit Error Rate (BER) across all SNR values due to its higher allocated power and simpler modulation scheme. User 2 (QPSK) shows moderate BER performance, while User 3 (16-QAM) exhibits the highest BER, as it uses a higher-order modulation scheme and is allocated lower power. This clearly demonstrates the trade-off between data rate and reliability in NOMA systems.

Figure 5.2 (Channel comparison for User 1) shows that Rician fading provides the best performance, followed by Nakagami and Rayleigh channels. The presence of a Line-of-Sight (LOS) component in the Rician channel significantly reduces fading effects, resulting in lower BER. Rayleigh fading, which represents a non-LOS scenario, shows comparatively higher BER, while Nakagami provides intermediate performance depending on its fading parameter.

From Figure 5.3 (All users across all channels), it is evident that BER decreases with increasing SNR for all users and channel types. However, users with better channel conditions and higher power allocation consistently outperform others. The results also highlight that channel conditions have a significant impact on system performance, with Rician channel consistently yielding lower BER values across users. Additionally, the performance gap between users increases at higher SNR values, emphasizing the importance of adaptive modulation and power allocation.

5.4 Evaluation of Quality Factors

In this section, the results are evaluated based on several quality factors, including:

- **Environment:** Efficient spectrum utilization allows multiple users to share the same frequency band, reducing network resource requirements.
- **Sustainability:** Adaptive modulation and dynamic power allocation improve long-term system efficiency.
- **Safety:** MATLAB-based simulations ensure minimal safety risks and no impact on physical infrastructure.
- **Ethics:** The project uses standard communication techniques and simulation data while maintaining academic integrity.
- **Cost:** Cost-effective performance improvements without additional hardware or bandwidth.
- **Type:** Simulation-based and analytical performance evaluation.
- **Standards:** Follows standard fading channels (Rayleigh, Rician, Nakagami) and AWGN models.

This section ensures that the obtained results meet expected performance requirements and suggests potential improvements for enhanced efficiency and reliability.

5.5 Summary

The chapter analyzed the performance of the proposed multi-user NOMA system with adaptive modulation and dynamic power allocation. The system's reliability was evaluated in terms of Bit Error Rate (BER) under different fading channels (Rayleigh, Rician, and Nakagami) and varying Signal-to-Noise Ratios (SNR). User-wise analysis showed that the far user with BPSK and the highest power allocation achieved the lowest BER, while the near user

with 16-QAM and the lowest power experienced the highest BER due to higher-order modulation and lower power. Channel comparisons highlighted that Rician fading slightly improved performance over Rayleigh, and Nakagami fading benefited closer users depending on the fading parameter. Increasing SNR reduced BER for all users, confirming the effectiveness of successive interference cancellation (SIC). Finally, the results were evaluated based on quality factors such as environment, sustainability, safety, ethics, cost, type, and standards, confirming that the proposed NOMA system improves spectral efficiency, fairness among users, and overall communication performance under realistic wireless channel conditions.

CHAPTER 6

Conclusions and Future Scope

6.1 Conclusions

This project presented the design and performance evaluation of a multi-user NOMA-based communication system integrated with Successive Interference Cancellation (SIC) and adaptive modulation techniques under different fading channel conditions. The system was successfully implemented and simulated in MATLAB to analyze real-time communication behavior and evaluate key performance metrics such as Bit Error Rate (BER).

From the results, it is evident that NOMA effectively enables multiple users to share the same spectral resources through power-domain multiplexing, thereby improving spectral efficiency compared to conventional orthogonal access methods. The implementation of SIC at the receiver plays a crucial role in separating superimposed signals, and its performance directly impacts the overall system reliability. The results demonstrated that improved detection and proper ordering in SIC significantly reduce error propagation, leading to better decoding accuracy.

The analysis of different modulation schemes showed that lower-order modulation techniques such as BPSK provide better BER performance due to their robustness, while higher-order schemes like QPSK and 16-QAM offer higher data rates at the cost of increased error probability. This highlights the importance of adaptive modulation, where the system dynamically selects the appropriate modulation scheme based on channel conditions to achieve an optimal balance between reliability and spectral efficiency.

Furthermore, the comparison of fading channels revealed that channel conditions have a significant impact on system performance. The Rician channel, due to the presence of a Line-of-Sight (LOS) component, provided the best BER performance, followed by Nakagami and Rayleigh channels.

This emphasizes the need for adaptive techniques to handle varying wireless environments effectively.

Overall, the project demonstrates that combining multi-user NOMA, SIC, adaptive modulation, and multi-fading channel analysis leads to significant improvements in system performance, including reduced BER, enhanced throughput, and better user fairness. These findings highlight the potential of NOMA-based systems as a key technology for next-generation wireless communication systems such as 5G and beyond.

6.2 Future Scope

The proposed system can be extended and enhanced in several ways:

- **Advanced Channel Models:** Extend the analysis to more realistic wireless channels, including correlated fading, shadowing, and time-varying conditions.
- **Multi-Antenna NOMA:** Implement MIMO-NOMA to further improve system capacity and reliability.
- **Machine Learning-Based Resource Allocation:** Use AI/ML techniques to optimize power allocation, user pairing, and adaptive modulation dynamically.
- **Hardware Implementation:** Test the proposed system on software-defined radios (SDRs) or FPGA platforms for real-time evaluation.

Mapping of Program Outcomes (POs) and Sustainable Development Goals (SDGs)

1. Contribution to Program Outcomes (POs) and Program Specific Outcomes (PSOs)

The project contributes to the attainment of the following Program Outcomes:

6.3 Mapping of Program Outcomes (POs)

PO1: Engineering Knowledge

This project applies fundamental concepts of *wireless communication, digital signal processing, and information theory*. It involves understanding *NOMA principles, adaptive modulation techniques, and fading channel models*, demonstrating strong engineering knowledge.

PO2: Problem Analysis

The project focuses on analyzing challenges such as *multi-user interference, error propagation in SIC, and channel variability*. Mathematical modeling and BER analysis are used to evaluate system performance under different conditions.

PO3: Design/Development of Solutions

An efficient NOMA-based communication system is designed using *adaptive modulation and dynamic power allocation*. The system improves spectral efficiency and reliability under various fading environments.

PO4: Conduct Investigations of Complex Problems

Extensive simulations are conducted to study system performance over *Rayleigh, Rician, and Nakagami fading channels*. Performance metrics like BER and SNR are analyzed to draw meaningful conclusions.

PO5: Engineering Tool Usage

The project utilizes *MATLAB* for modeling, simulation, and performance evaluation, demonstrating the effective use of modern engineering tools.

PO6: The Engineer and The world

The proposed communication system contributes to improving *wireless connectivity and spectrum efficiency*, which are essential for modern communication systems and societal development.

PO7: Ethics

By improving *spectral efficiency and power utilization*, the system supports energy-efficient communication, indirectly contributing to sustainable technological development.

PO8: Individual and Collaborative Team work

The project follows ethical engineering practices by ensuring *accurate simulation, proper citation of research work, and unbiased performance evaluation*.

PO9: Communication

The project involves effective collaboration, task sharing, and coordination among team members during *design, simulation, and analysis phases*.

PO10: Project Management and Finance

Technical results are clearly presented through *graphs, reports, and documentation*, ensuring effective communication of complex engineering concepts.

PO11: Life-Long Learning

The project is planned and executed efficiently by managing *time, computational resources, and project milestones*, demonstrating basic project management skills.

2. Program Specific Outcomes (PSOs)**PSO2: Demonstrate technical competency in communications and signal processing**

The project demonstrates technical competency in *multi-user NOMA, modulation/demodulation techniques, and BER analysis* using appropriate tools and simulation techniques, fulfilling PSO2.

3. Alignment with Sustainable Development Goals (SDGs)

The project supports the following Sustainable Development Goals:

SDG 7: Affordable and Clean Energy

The project promotes energy-efficient wireless communication by optimizing power allocation for multiple users, reducing unnecessary energy consumption in transmission.

SDG 9: Industry, Innovation and Infrastructure

By exploring advanced NOMA communication techniques, the project contributes to innovative and sustainable communication infrastructure development.

SDG 12: Responsible Consumption and Production

The project optimizes spectral and power resources in wireless systems, promoting responsible use of technological and energy resources.

Overall Impact

The project not only fulfills academic learning outcomes but also contributes toward broader societal and environmental objectives. By integrating technical knowledge with sustainability considerations, the work demonstrates alignment between institutional Program Outcomes and global development priorities.

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